

PERSONAL DETAILS

Jerusalem, Israel
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EDUCATION

M.Sc. Genomics & Bioinformatics, Biotechnology

Hebrew University | 2024 – 2026
Coursework completed. Thesis submission pending.

Recipient of the Faculty Excellence Scholarship (2024–2026). Collaborating with Hadassah Medical Center on clinical IRD cohort analysis. Includes a structured industry practicum at Pre-Cure Biotech (150 hrs, 10 cr).

B.Sc. Marine Biotechnology

Ruppin Academic Center | 2021 – 2024
GPA: 91/100

Conducted a year-long research project on octopus motor learning and neural plasticity.

LANGUAGES

Hebrew	Native
English	Fluent
Arabic	Basic

BEYOND THE LAB

Israeli National Thai Boxing Champion (2012–2017): 4x National Champion. Developed elite self-discipline, resilience, and performance under pressure.

Certified Skipper (licenses 12, 13, 30) and Diver (2 stars).

Shalev Yaacov

M.Sc. Genomics & Bioinformatics, Biotechnology program, at the Hebrew University of Jerusalem, Prof. Yuval Tabach lab.

Specializing in ML-driven pipelines that integrate Multi-Omics and clinical data for disease gene discovery and precision medicine applications.

Awarded the Faculty of Biotechnology Excellence Scholarship (2024–26). Holds a B.Sc. in Marine Biotechnology with an internship in Marine Agriculture, graduated with GPA of 91.

RESEARCH

M.Sc. Research | Prof. Yuval Tabach Lab

2024–2026

Partitions 450+ Inherited Retinal Disease (IRD) genes into co-evolved functional modules using Normalized Phylogenetic Profiling (NPP) across ~2,000 genomes and HPO-based phenotypic clustering.

Predictive Framework: Built a clinical decision-support system for causal gene prediction, integrating multi-omics data into a bidirectional phenotype-to-gene inference engine.

Novel Gene Discovery: Built and open-sourced a multi-modal deep learning model (PyTorch) ranking novel IRD gene candidates genome-wide (~7.5× random baseline), validated on 4,254+ real-world clinical cases in collaboration with Hadassah Medical Center.

Protein Engineering | Miruku R&D Collab. | Prof. Oded Shoseyov Lab

2025

Collaborated with Miruku's R&D team on plant-based recombinant protein production; executed expression, purification, and characterization of Lactoferrin and Keratin in bacterial and plant expression systems.

Co-authored "Plant molecular farming to produce milk components" in *Alternative Dairy Products and Technologies* (Academic Press, 2026).

B.Sc. Project – Octopus Motor Learning | Dr. Nir Nesher Lab

2024

Designed and analyzed motor learning experiments in octopuses, investigating variability in adaptive learning strategies — flexible strategy-based adaptation vs. fixed tactical sequences — with implications for neural plasticity and behavioral resilience.

INDUSTRY PLACEMENT

Pre-Cure Biotech | MSc Industry Practicum (Forward Deployed Engineering)

PRESENT

Designing and cloud-deploying a relational data management system for a personalized oncology startup, translating complex biological workflows into structured, queryable data infrastructure; developing a Python backend for analytics and automation, and an AI-assisted layer (incl. OCR) for experimental data interpretation.

COMPUTATIONAL & LAB SKILLS

Coding & Bioinformatics: Python (Pandas, NumPy, Scikit-learn, PyTorch, Matplotlib, OCR), R (statistics and visualization), SQL, Linux/Bash, Git/GitHub, HPC/SLURM. Multi-omics data integration, biological network analysis, high-throughput genomic data processing, and pipeline development.

ML Frameworks: Bayesian inference and probabilistic modeling, ensemble methods (XGBoost, Random Forest), deep learning (PyTorch, multi-modal fusion architectures, pretrained language models), graph-based unsupervised clustering and community detection, model evaluation and validation.

AI-Integrated Workflows: Research-grade use of AI coding agents (incl. Claude Code, Cursor, Codex, Google Labs) for pipeline development and scientific reasoning; custom agent and prompt engineering; local LLM inference (Ollama).

Molecular Biology (Wet-Lab): Expression of bovine Lactoferrin & human Keratin in E. coli and Arabidopsis; proficient in transgenic plant selection, transformation, FPLC purification, cloning, qPCR, and ELISA.

Tools & Databases: Clustal Omega, BLAST, PyMOL, PDB, AlphaFold, ESM2, Rummagene, FUMA, EnrichR, Cytoscape, STRING, UniProt, GeneCards, OMIM, ClinVar, gnomAD, UCSC Genome Browser, NCBI, Ensembl, Seurat, Scanpy.

MILITARY SERVICE

Squad Commander & Platoon Sgt | IDF Combat Engineering

2017–2020

Led combat teams in field operations and managed the battalion operations room, oversaw the training and command of new recruits. Certified Heavy Engineering Equipment Commander (09) and Combat Pioneer Commander (08).

VOLUNTEER EXPERIENCE

United Hatzalah & Jerusalem Veterinary Center

2021–2025

Monitored vitals and provided companionship for the elderly via the "Ten Kavod" project (~140 hrs/year). Assisted in veterinary procedures and emergency cases, gaining hands-on experience in clinical management.